The Data Science Workflow

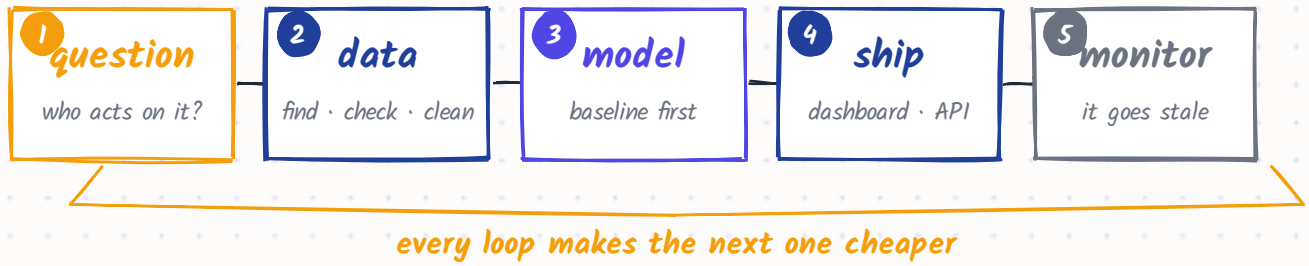
Lesson 2 · five steps, and three ways projects die

In this lesson → ① Why projects die ② The loop ③ The question ④ The data ⑤ Baseline ⑥ Ship ⑦ Monitor ⑧ Interview Q&A

Module dataset: ChurnLens, a deliberately messy telecom CSV we clean, explore and model across Module 1.

1 The loop

> Every data project is the same five steps, run as a **loop**: question → data → model → ship → monitor. What you learn at step 5 rewrites step 1.



2 Step 1 · The question

→ **Who** acts on this answer?
→ **What** will they do differently?
→ **How** will we know it worked?

✗ Vague

- "Can we do something with churn data?"
- "Build a model for the customer team"

✓ Decision-shaped

✓ "Which 200 customers should retention call on Monday?"
✓ "Which 3 features predict a refund?"

Write it down
One sentence, agreed with the person who will act. If nobody can answer the three questions, the project is already dead.

3 Step 2 · The data

→ Where does it live — warehouse, app database, an export someone emails?

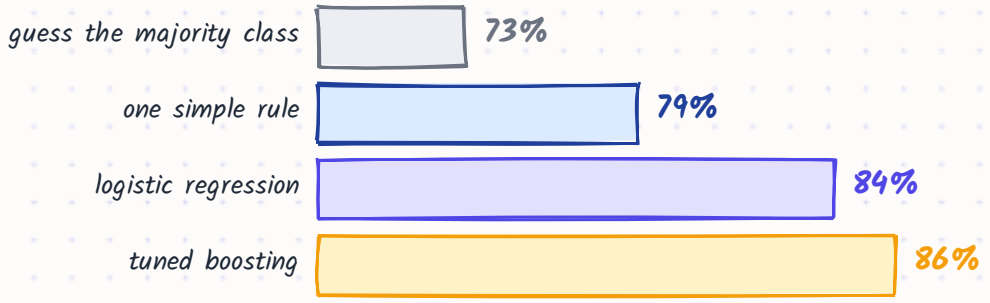
→ Is it complete — row counts, date ranges, duplicates, missing values?

→ **When does it arrive?** A prediction that needs data which lands a week late cannot be acted on.

Leakage

A column that only exists *after* the outcome. `cancellation_date` predicts churn perfectly and teaches you nothing. Ask of every feature: **would I have this at prediction time?**

4 Step 3 · Baseline first



ship the simple one, then argue for the extra 2%

Baseline	What it is	Why it matters
Majority class	always predict the most common label	the floor — beat this or stop
One rule	e.g. no login for 30 days → churn	often 80% of the value, in an afternoon
Simple model	logistic regression, default settings	the number your fancy model must beat

Say this

A model without a baseline is a number with no meaning.

5 Steps 4 & 5 · Ship and monitor

Shipping can be

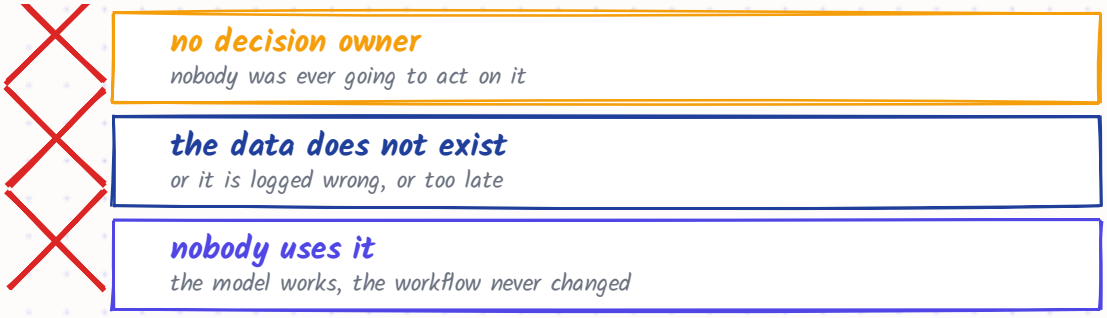
- ✓ a dashboard opened on Monday
- ✓ a weekly CSV or email
- ✓ an API endpoint
- ✓ a report with one recommendation

Monitor

- → inputs: arriving? same shape?
- → outcomes: still right?
- → a review date in the calendar
- → a named owner

> **Rule**
> Pick the smallest thing that changes the decision. A CSV people use beats an endpoint nobody calls.

6 The 3 places projects die



Death	Symptom	Prevented by
No decision owner	"Interesting!" — and nothing changes	naming the person in step 1
Data does not exist	the key column was never logged, or lands too late	checking timeliness before modelling
Nobody uses it	the model works; the workflow never changed	shipping into an existing habit

7 Interview corner

Q. How would you approach a new data science project?

A. Agree the decision and its owner → check the data exists and arrives in time → build a baseline → then a model → ship the smallest useful thing → monitor it.

Q. What is data leakage, in one line?

A. Using information at training time that would not be available at prediction time — it inflates scores and collapses in production.

Q. Why start with a baseline?

A. It turns a score into a decision: you can say what the model buys you over the obvious answer.

Q. What would you do differently on a past project?

A. Define what "shipped" means on day one, with a named owner.

⚡ Revise in 60 seconds

- → 5 steps: question → data → model → ship → monitor
- → step 1 names a decision AND a person
- → leakage = a column that exists only after the outcome
- → always beat a baseline before tuning
- → ship the smallest thing that changes the decision
- → 3 deaths: no owner · no data · no adoption

Project checklist — copy this into every new project

- ✓ the decision is written in one sentence
- ✓ the person who will act is named
- ✓ success is defined before modelling
- ✓ every feature exists at prediction time
- ✓ the data arrives early enough to act on
- ✓ a baseline score is written down
- ✓ the simplest useful output is chosen
- ✓ someone owns it after I move on
- ✓ a review date is in the calendar
- ✓ inputs and outcomes are monitored

Next lesson → DS-03
Lab: **NumPy** — arrays, vectorisation, and why loops are slow
Watch on YouTube: **@TheCodeNotebook** · Notes: **thecodenotebook.com**